

Outline:

1. Syllabus + Evaluation Method
2. Integrated Development Environment (IDE) vs ...
3. Platforms for versions control (GitHub or GitLab)
4. Compiler
5. Priority Of Arithmetic Operations (Mathematically ...)
6. My First Program + Environment Explanation
7. Summation Program.
8. Finding the Maximum Value BTW (2 and 3) Variables.

Lab 1: Introduction to C++ & Basic I/O

Lab Objectives

By the end of this lab, students should be able to:

1. Understand what a C++ program looks like (structure).
 2. Compile and run a simple C++ program.
 3. Use basic input/output (cout, cin).
 4. Understand variables and simple data types
- What is a compiler?

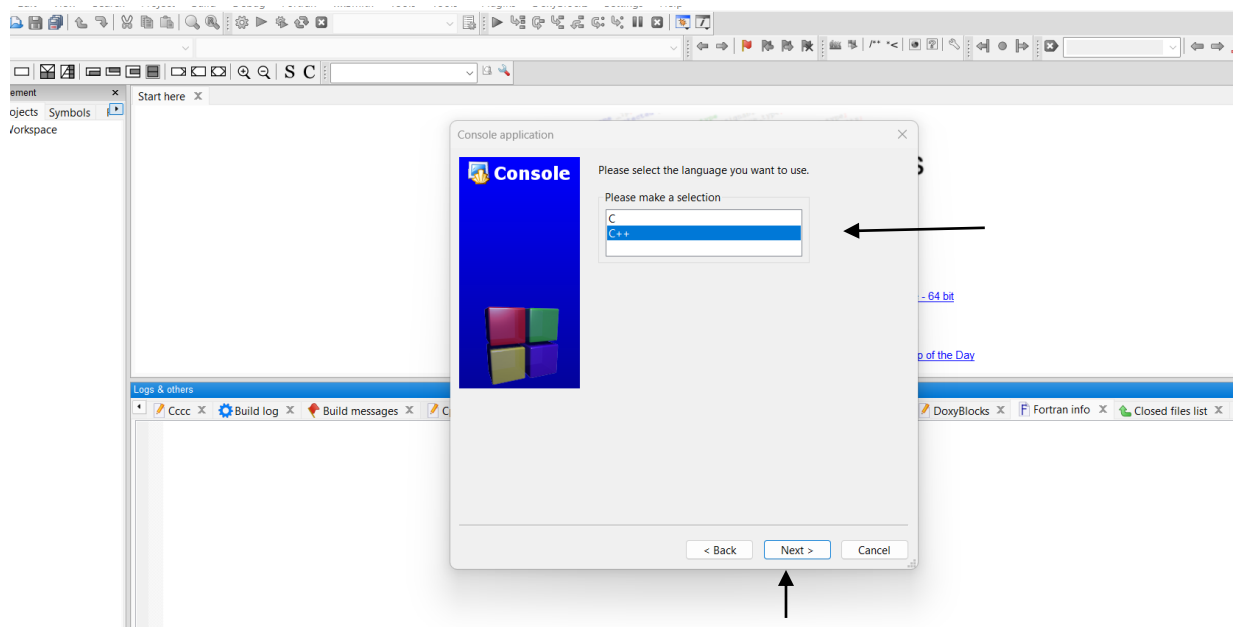
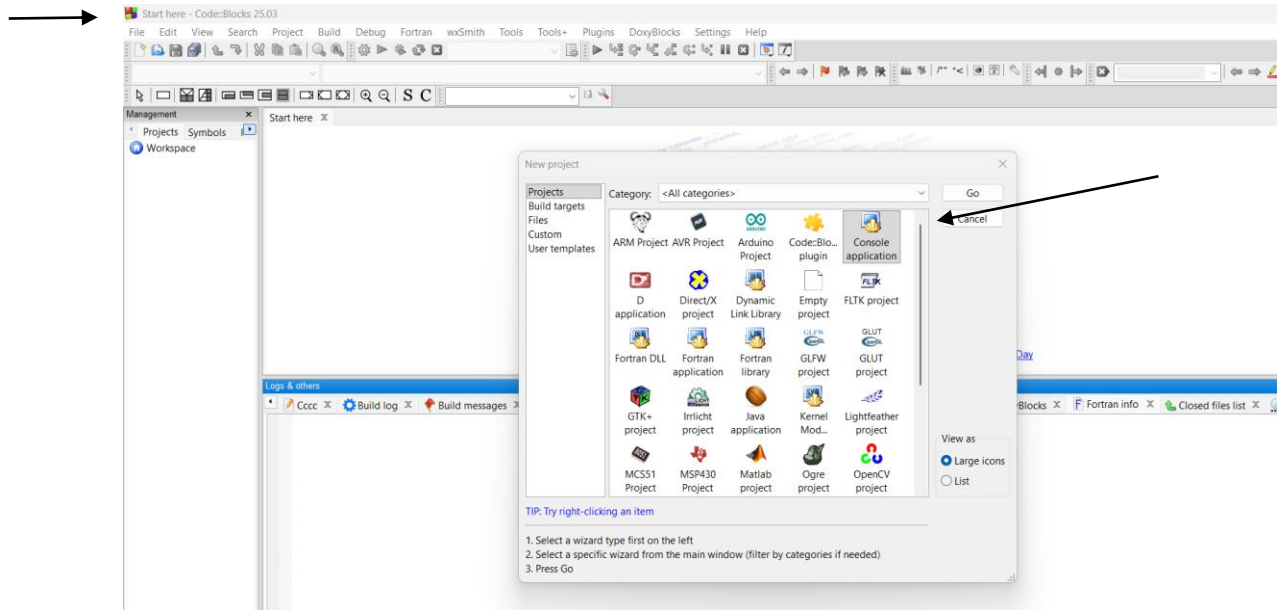
- What is IDE? (Recommend: CodeBlocks, Visual Studio Code + MinGW or use online platform)

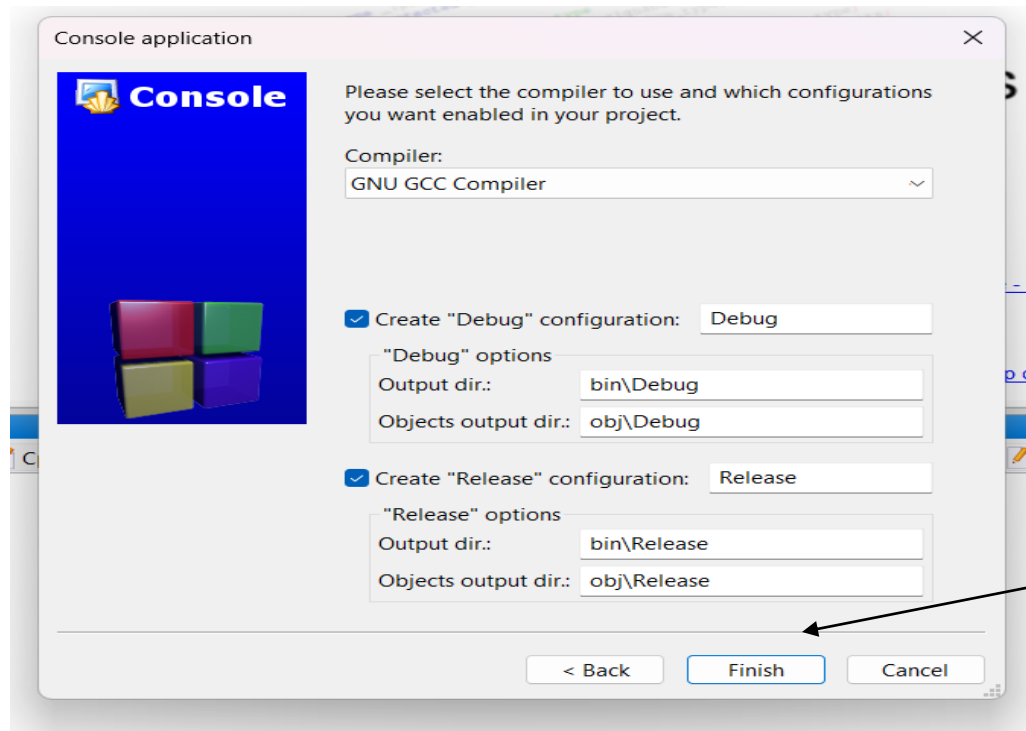
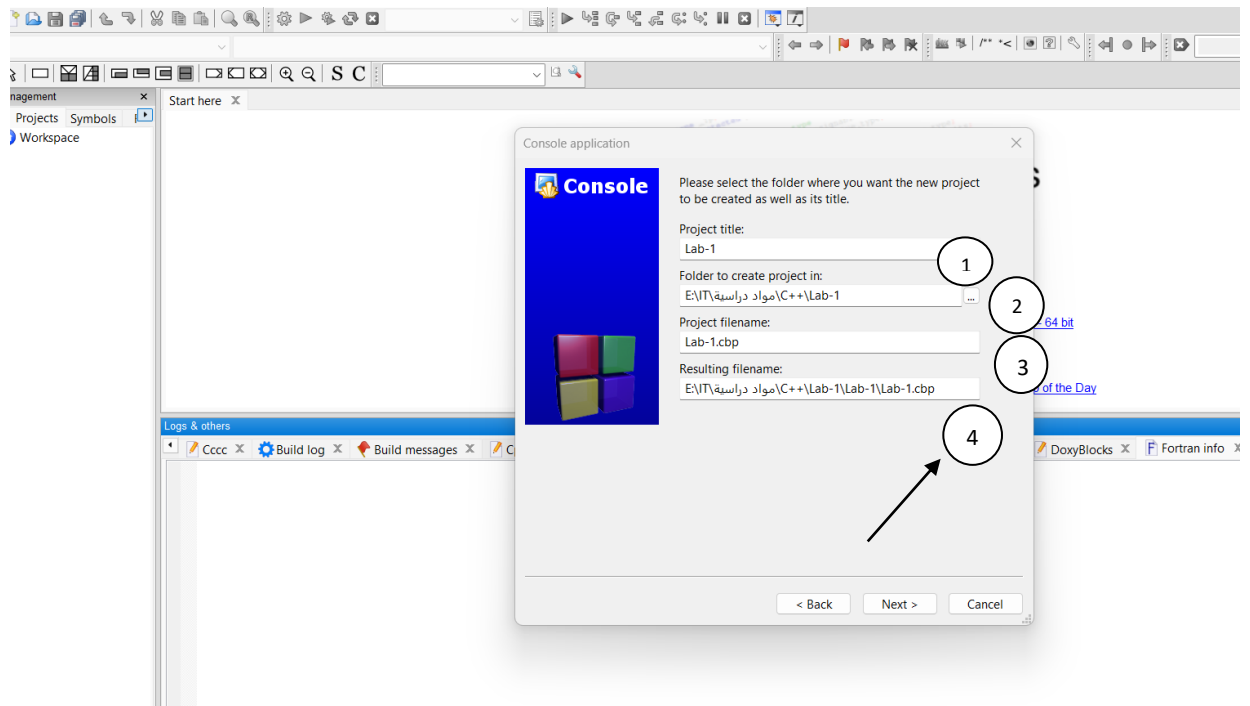
A- Open Code::Blocks

B- Create a New Project

1. Click File → New → Project
2. Select Console Application
3. Choose C++

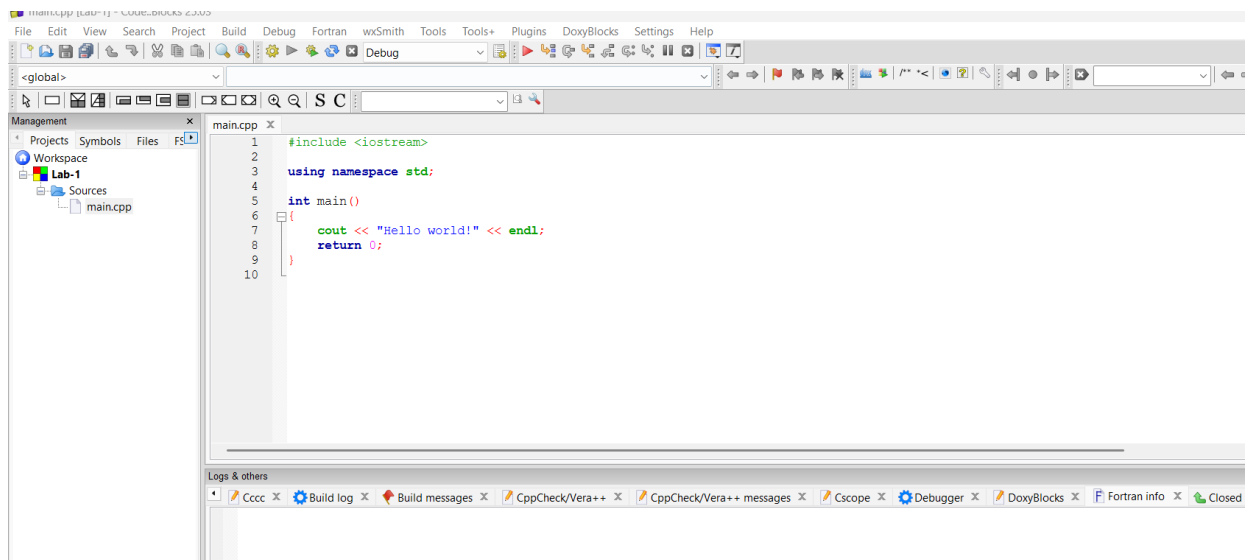
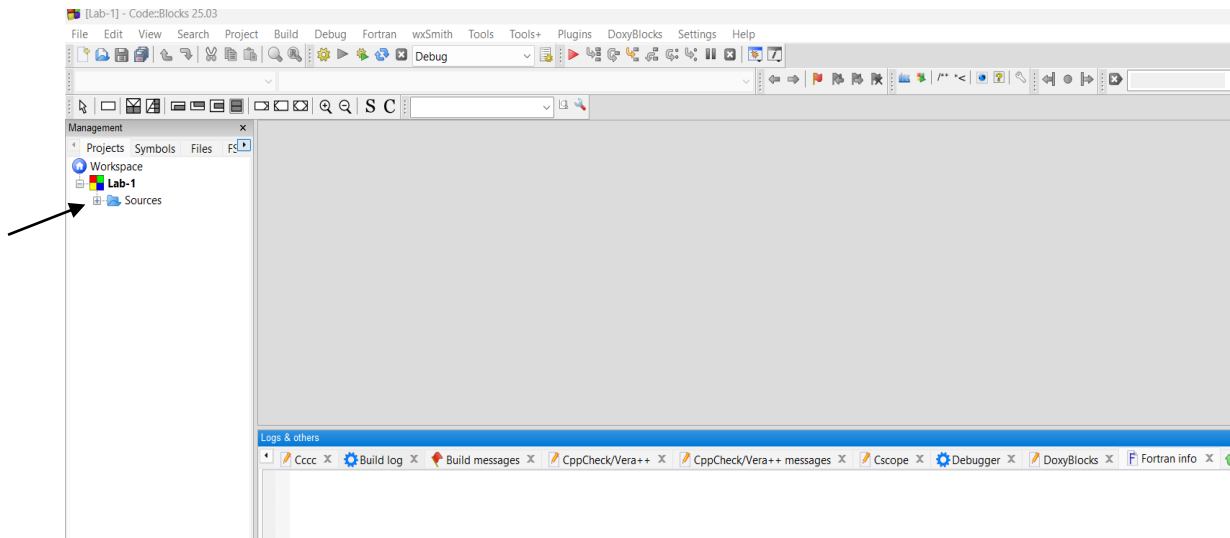
4. Click Next





C- Set Project Name

1. Enter a project name (example: *Lab1*).
2. Choose where to save it.
3. Click Next → then Finish



D- Open the main.cpp File

You will see:

- On the left, under Projects, expand your project.

- Expand Sources.
- Double-click main.cpp this is where you write your C++ code.

E- Delete the default code (optional)

Code::Blocks creates a sample program.
You can remove it and paste your code:

F- Build (Compile) the Program

Click: Build (or press F9) This compiles your program.

If there are no errors, you will see:

✓ Build succeeded

G- Run the Program

Click: ► Run or press Ctrl + F10

A black console window will open showing your output.

H- Build & Run Together (Most Useful)

Click: Build and Run or press F9 (in some versions)

This compiles and runs automatically.

- What is a source code file (.cpp)?

Program Structure Overview

```
#include <iostream>
using namespace std;
int main() {
    cout << "Hello, World!" << endl;
    return 0;
}
```

Task 1: Write a program that show the following message

The Expected Output:

Welcome to C++ programming!

20/11/2025

```
#include <iostream>
using namespace std;
int main () {
    cout << "Welcome to C++ programming!";
    cout<<endl;
    cout<<"20/11/2025";
    return 0;
}
```

Variables & Data Types

Int, float, char, string, bool.

```
int age = 20;
```

```
float gpa = 3.7;
```

```
char grade = 'A';
```

```
string name = "Ali";
```

```
bool isAStudent = true;
```

....

Task 2: Input from the user

```
#include <iostream>
using namespace std;
int main() {
    string name;
    int address;
    cout << "Enter your address: ";
    cin >> name;
    cout << "Enter your address: ";
    cin >> address;
    cout << "Hello " << name << ", your address is " << address << "\n";
    return 0;
}
```

Task 3: Simple arithmetic

write a program that:

- i. Reads two numbers
- ii. Prints the sum, product, difference

```
#include <iostream>
using namespace std;
int main() {
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << "Sum = " << a + b << endl;
    cout << "Product = " << a * b << endl;
    cout << "Difference = " << a - b << endl;
    return 0;
}
```

Mini-Quiz / Check Understanding

- What is the purpose of main ()?
- Difference between cout and cin?
- What is a variable?
- What is int vs float?

Lab Assignment (Homework)

Write a C++ program that:

1. Asks the user for:
 1. Their name
 2. Their department
 3. Their age

2. Prints a formatted message:

Example expected output:

```
Welcome Ali
Department: Artificial Intelligence
Age: 19
```

IF Condition + Comparison Programs

```
if (condition) {
    // code to execute if condition is true
}
```

Example of a condition:

```
✓ A > B
✓ x == 10
✓ grade != 'A'
✓ age >= 18
```

Example 1: Check if a number is positive

```
int x;
cout << "Enter a number: ";
cin >> x;

if (x > 0) {
    cout << "The number is positive.";
}
```

Example 2: Write a program that Compare Two Numbers

```
#include <iostream>
using namespace std;

int main() {
    int A, B;

    cout << "Enter two numbers: ";
    cin >> A >> B;

    if (A > B)
```

```
    cout << "A is greater than B";  
    else if (A < B)  
        cout << "B is greater than A";  
    else  
        cout << "Both numbers are equal";  
  
    return 0;  
}
```

ToDo:

Example 3: Write a program that Compare Three Numbers

You need to create GitHub account and join the course repository.

<https://github.com/Mohammed20201991/Programming-UF-Basics.git>

CodeBlocks Installation



